

1. THE PROBLEM, AND WHAT MAKES IT HARD

We predict `energy_usage` for a campus building from its identity, time, weather, occupancy and its preceding reading. The headline difficulty is not finding signal — `previous_usage` alone correlates 0.945 with the target, and simply copying it forward scores RMSE 6.40. The difficulty is that *the test set is systematically harder than the training set*, in a way that silently corrupts how a team measures its own progress.

2. TWO FINDINGS THAT SHAPED EVERYTHING

(a) **This is not a time series, despite looking like one.** The data has `hour`, `day_of_week` and `month` columns, a lag feature, and a tidy ID split (train EC000001–008000, test EC008001–011000). The natural response is a time-aware split. We tested that assumption instead of inheriting it: there is no timestamp, so ordering is undefined; every correlation between ID number and any feature falls within ± 0.016 ; and an ID-ordered holdout scored **3.606** against **3.560** for random 5-fold — a gap smaller than the ± 0.071 fold noise. We therefore use random K-fold, keeping 20% more training data than a time split would allow, with evidence rather than assertion behind the choice.

(b) **Standard validation is 11.2% optimistic here.**

Every gap-prone feature is missing 4–5× more often in test than in train. 18.2% of test rows carry at least one gap versus 5.8% of training rows; for rows with two or more gaps the split is **7.7% versus 0.14%** — **a 55× difference**. A model validated on training-like data is therefore graded on an easier problem than it will face. We measured the bias directly: one fitted model, same validation rows, scored **3.560** as-is and **3.960** once the fold was degraded to test-level missingness.

The consequence is worse than an inflated number. Models degrade at different rates, so naive validation can *misrank* them. On our own earlier candidates, a Ridge model appeared 0.20 RMSE better than gradient boosting and was actually 0.08 *worse* under test conditions. All model selection in this work is therefore made on **RMSE with test-level missingness injected into every validation fold**, fixed before any model was tuned.

3. APPROACH

Features. Exploration showed buildings differ in *shape*, not merely in level: mean usage spans 30.8 (Administration) to 79.2 (Science), while variability ranges from sd 6.5 (Residential, highly regular) to 19.1 (Lecture Halls, volatile). We encode this explicitly — per-building intercept, 24-hour profile, weekend offset, and per-building slopes on occupancy, temperature and previous usage (366 features) — plus cyclical hour encoding and per-column missingness indicators.

Imputation. This proved the single most valuable decision. A median is not merely imprecise: the building slopes are products `is_building × value`, so one missing input collapses the entire building-specific relationship for that row. We instead train a dedicated LightGBM per gap-prone column, predicting it from building, hour and the remaining drivers, then regenerate the interaction terms. Tested against the alternatives on identical folds:

IMPUTATION STRATEGY	RMSE
Global median (standard practice)	3.852
Building-level median	3.614
MICE / iterative imputer	3.562
Sub-model (LightGBM per column)	3.339

Augmentation. Training contains roughly 11 multi-gap rows against ~231 in test, so the model has almost no exposure to the regime it is graded on. Degrading the training data to test-level missingness improved every

model tested (CatBoost by 0.265) and halved fold variance. This uses only the observed missingness *rate* of the test features — no test labels are involved.

4. MODEL SELECTION, AND REJECTING THE ENSEMBLE

Ridge with explicit building interactions outperformed every gradient-boosted alternative. The structure here is close to additive — a building's base level, its own daily profile, and near-linear responses to its drivers — which a regularised linear model fits directly and extrapolates from sensibly, while trees must approximate smooth relationships with axis-aligned splits.

MODEL (REALISTIC VALIDATION)	RMSE	MAE
Ridge + building interactions	3.233	2.528
CatBoost	3.487	2.708
LightGBM	3.548	2.759
XGBoost	3.582	2.783
4-model weighted stack (nested)	3.261	—

We built the ensemble and then rejected it. Scored conventionally it appeared best (3.226), but those blend weights had been fitted on the very predictions being scored. Under nested validation — weights learned on inner folds, evaluated on an untouched outer fold — it scored **3.261**, **worse than the single Ridge at 3.233**. Its apparent advantage was 0.035 RMSE of self-deception: the same category of error we had diagnosed in the validation protocol, caught this time in our own work.

5. RESULT

3.233 RMSE	2.528 MAE	0.969 R²
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Against references of 18.45 (predict the mean), 13.22 (building mean) and 6.40 (copy `previous_usage`, no model). All figures are under injected test-level missingness; conventional validation reports materially better numbers we do not consider trustworthy.

6. ERROR ANALYSIS

Residuals are centred and symmetric, with no bias across the prediction range. Error tracks each building's intrinsic volatility: Lecture Halls hardest, Residential easiest. Much of this is irreducible — a lecture hall genuinely is less predictable than a dormitory — but it means a campus-wide average understates uncertainty on exactly the buildings an operator would most want warning about. Error also rises with the number of missing inputs, though far more gently than median imputation produces.

7. LIMITATIONS AND NEXT STEPS

- **No timestamps**, so genuine lag or rolling features are impossible; `previous_usage` is the only history available.
- **Imputation error is invisible downstream.** Missingness flags let the model discount estimated values; propagating imputation uncertainty would be the principled fix.
- **Augmentation rates are calibrated to the public test set.** If the private set is harsher those rates are slightly off, though this approach should still degrade more gracefully than median-based alternatives.
- **High-variance buildings dominate residual error.** Per-building or mixed-effects models are the next step, alongside a cost-weighted error view — a 5-unit error on a 30-unit admin block matters more proportionally than on an 80-unit science building.
- Eight further variations were tested and rejected on evidence — log and delta target framings, extra interactions, splines, imputer tuning, and a specialist model routing rows lacking `previous_usage`. None improved on the reported figure, which we read as a sign the remaining error is largely irreducible.